
Predicting if passengers will survive the titanic or not

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1. Introduction

In this project I used the **Titanic dataset**, which contains 890 Titanic passengers' info. Each row includes their age, sex, class, and fare. My goal was to predict which passenger has survived and what the common traits of a survivor of the Titanic are.

2. Cleaning Summary

The raw dataset had some problems that needed fixing before analysis:

- **Missing values:** I found two columns that had some missing values, one of them, the "Cabin" column was mostly missing, so I completely dropped that column. The other one, "Age" had fewer missing values, it had approximately 20% missing, and I dropped them because its hard to predict and fill the ages of passengers.

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3. Models used

I tried three models in this project, LogisticRegression, RandomForest, and KNN, and then I compared their f1 scores to see the best model, and these were the results

- 1-RandomForest(0.61)
- 2-LogisticRegression(0.58)
- 3-KNN(0.46)

4. Neural Network

After that I made some neural network using ReLu and also softmax / sigmoid, there were 3 Dense layers in each, and also i added some Dropout layers. after 10 epochs I tested the difference in accuracy, and i found that with the dropout layers accuracy would drop and be less than without the dropout layers

5. Results and Comparison

After training the models and making the neural networks, I compared their F1 score to check which way is best, and I found some surprising results! And the results are as follows:

- 1 - RandomForest (0.61)
- 2 - Logistic Regression (0.58)
- 3 - Sigmoid + Dropout (0.53)
- 4 - KNN (0.46)
- 5 - Softmax + Dropout (0.25)

6. What I learned

I learned that neural networks are not always the best to use for predections.

7. What I Would Do Next

For future work, I will try to fill the "Age" column, and also I would try more models to see if there is a better one, and also I would try to put more layers in the neural networks and maybe the optimal number of dropout layers.
